

Systems Thinking and Design

Phase 2 Report: The Breakfast Paradox at IIIT Hyderabad

Invictus

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1 Introduction

Phase 1 established what the breakfast mess system is and how it operates. Phase 2 moves from description to explanation: building a holistic map of the system, identifying the structures and mental models producing the problem, and reframing the operational problem as a systemic one.

Phase 2 covers three activities and five deliverables. This report presents all five in order.

#	Deliverable	Location in this report
1	System Map	Section 2.1 (Activity 6)
2	Causal Loop / Feedback Analysis	Section 2.2 (Activity 6)
3	Systemic Problem Analysis	Section 3 (Activity 7)
4	Reframed Systemic Problem Statement	Section 4.1 (Activity 8)
5	Design Opportunity Statement	Section 4.2 (Activity 8)

Evidence base and confidence convention

Phase 2 added three primary sources beyond Phase 1's student interviews: an interview with the Campus Facilities Services (CFS) Chair, a walkthrough of the Kadamba kitchen, and four Campus Dining Services (CDS) posters. Every claim in this report is marked **confirmed** (a named source stated it directly, or two sources corroborate it), **single sourced** (one account, not cross-checked), **candidate** (a plausible named mechanism, not verified), or **assumed** (a systems-thinking inference, stated as such and never given the weight of fact).

Correction to Phase 1's picture

Three Phase 1 findings were corrected by the new evidence. Governance is Campus Dining Services under Campus Facilities Services, a CDS Chair, a faculty CDS Committee with final say, a CDS Student Council doing first review, and one admin per dining hall, not the "Mess Office / Mess Committee" structure assumed earlier; whether these are the same bodies under different names is unresolved and left open. Bakul and Palash are not a converted warehouse pair: Palash is under construction and cooking nothing on site, with both served by one outside vendor until a shared kitchen opens around December 2026. And Skip Meal does reach the kitchen, same-week declarations adjust cooking quantity, though not the four-day ingredient order.

2 Activity 6: Develop a Rich Picture / System Map

Purpose: build a holistic representation of the system and its interconnected elements.

2.1 Deliverable 1: System Map

Deliverable 1: System Map: actors, processes, information, resources, constraints, conflicts, dependencies, formal and informal structures, bottlenecks, tensions and feedback loops

Actors, process, resources and constraints

Students are the registrant, the billed party and the consumer. Interns require a mandatory Food Card; faculty and staff may hold one, billed against salary. CDS sits under CFS with the structure described above; catering is outsourced by open tender. Academic administration sets the class schedule and holds no formal role in mess governance. Students separately run an informal resale market through a WhatsApp group known as Mess Cell.

Three groups were not mapped in Phase 1 and are added here. Faculty and early-shift ground, security and housekeeping staff are a real but uncounted demand stream on the same food supply (assumed). The Canine Council, a student body managing campus dogs, plausibly absorbs kitchen waste informally (assumed). Vendors and suppliers upstream of the kitchen are part of the system rather than an opaque box.

Registration runs monthly and locks four days ahead; billing follows registration count, not attendance. That four-day lock governs ingredient ordering, so nothing about a specific morning can change what is bought. A later stage is more flexible: same-week Skip Meal declarations do adjust how much of the ordered stock is cooked (confirmed). A weekly consumption-driven stock check sits between the two. Turnout by category is confirmed: high-turnover items above ninety percent, general items around seventy, breakfast specifically thirty five to forty five. Service runs 7:30–9:30 AM, with roughly eighty percent of students arriving near 9:20, entry closing at 9:30 and refills continuing to 10:00.

A student who never registers is still auto-assigned a vegetarian meal somewhere. Exemptions exist for internships, PhD students and long-term medical need, though the monthly infrastructure fee still applies. Walk-in access is deliberately throttled near the end of a billing cycle to protect food for registered diners.

Dining Hall	Capacity	Cuisine	Dining Type	Kitchen
Kadamba	1200	South Indian	Mixed	On site, automated
Bakul	750	North Indian	Mixed	Off site, shared with Palash
Palash	450	North Indian	Vegetarian	Off site, under construction
Yuktahar	350	Satvik and Jain	Vegetarian	On site, manual

Item	Student rate	Card rate	Walk-in rate
Veg Breakfast	48	55	81
Double Egg Breakfast	66	76	111
Veg Lunch	65	75	111

Item	Student rate	Card rate	Walk-in rate
Veg Dinner	65	75	111
Chicken Biryani Lunch	155	178	263

Prices in rupees. The student rate applies only to meals registered in advance; anyone unregistered pays the walk-in rate, seventy to ninety percent higher, a standing financial nudge toward registering under uncertainty.

An unresolved discrepancy is recorded rather than smoothed over: the CDS poster lists Kadamba's capacity as 1200, while a separate interview gives 420 seats against a stated 1,200 turnout with a 1:3 ratio maintained deliberately. These are probably registration capacity and physical seats respectively, but that is a guess.

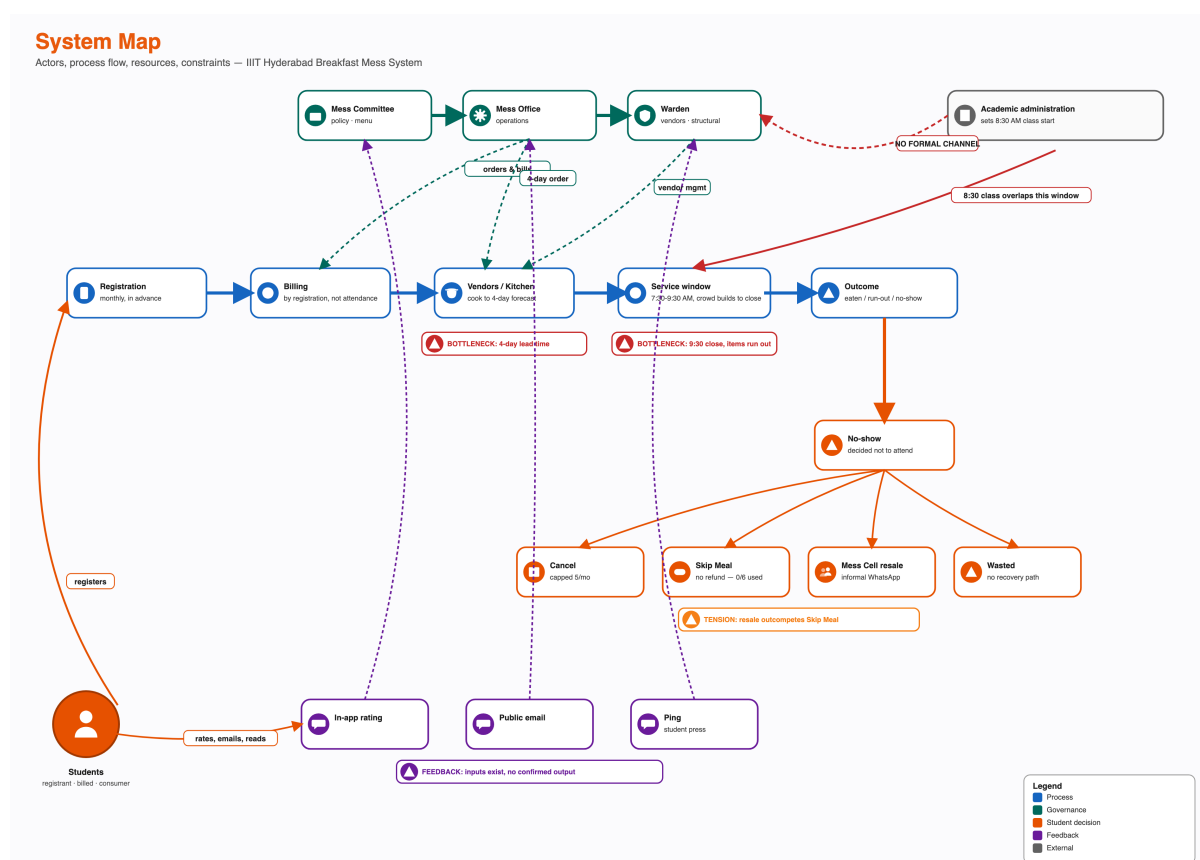


Figure 1: System Map, actors, process flow, resources and constraints, with bottlenecks, tensions and feedback channels marked.

Three kitchens, not one

Kadamba runs automated preparation and portions into containers out of sight. Yuktahar cooks manually and live, and runs separate Jain and pure vegetarian lines no other hall offers. Palash cooks nothing on site; it and Bakul are served by one outside vendor. These are three structurally different systems, so a finding about turnout, waste or quality at one should not be assumed to transfer to the others.

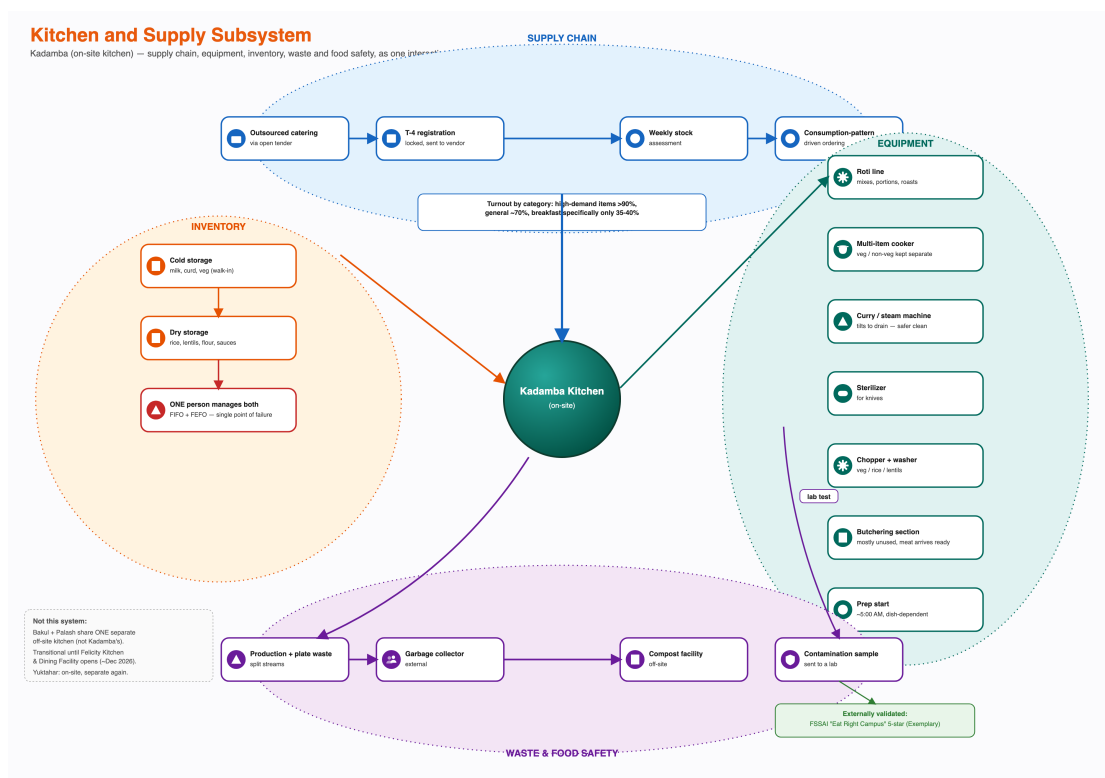


Figure 2: Kitchen and supply subsystem behind the vendor/kitchen node.

Formal and informal structure

Low-intensity feedback goes directly to the CFS Head; high-intensity feedback escalates through the CDS Student Council to the CDS Committee. That pathway has produced a real result: a standing complaint about a single semester-long menu escalated and returned as a menu rotating every two weeks, the clearest working feedback loop in this project.

Outside the formal structure sit Mess Cell, Ping (the student publication, covering mess issues with no confirmed authority to act), and at least one unofficial registration-management tool. Academic administration has no formal or informal relationship to any part of mess governance, a confirmed absence, not a research gap.

Conflicts, dependencies and bottlenecks

Students depend on the vendor for food; the vendor depends on no individual student. The vendor depends on CDS for a four-day forecast. CDS depends on students registering accurately, but nothing in billing requires that accuracy, the one relationship where the party with less to lose controls an input the other cannot replace. Kadamba's entire inventory depends on one person with no named backup.

Four bottlenecks are confirmed: the four-day ordering lock, which no same-day signal can reach; the 9:30 close against a crowd cresting just before it; the absence of any channel between Academic administration and mess governance; and waste never being measured by volume, so it cannot function as a signal for anything.

2.2 Deliverable 2: Causal Loop / Feedback Analysis

Deliverable 2: Causal Loop / Feedback Analysis, reinforcing and balancing loops tested for closure

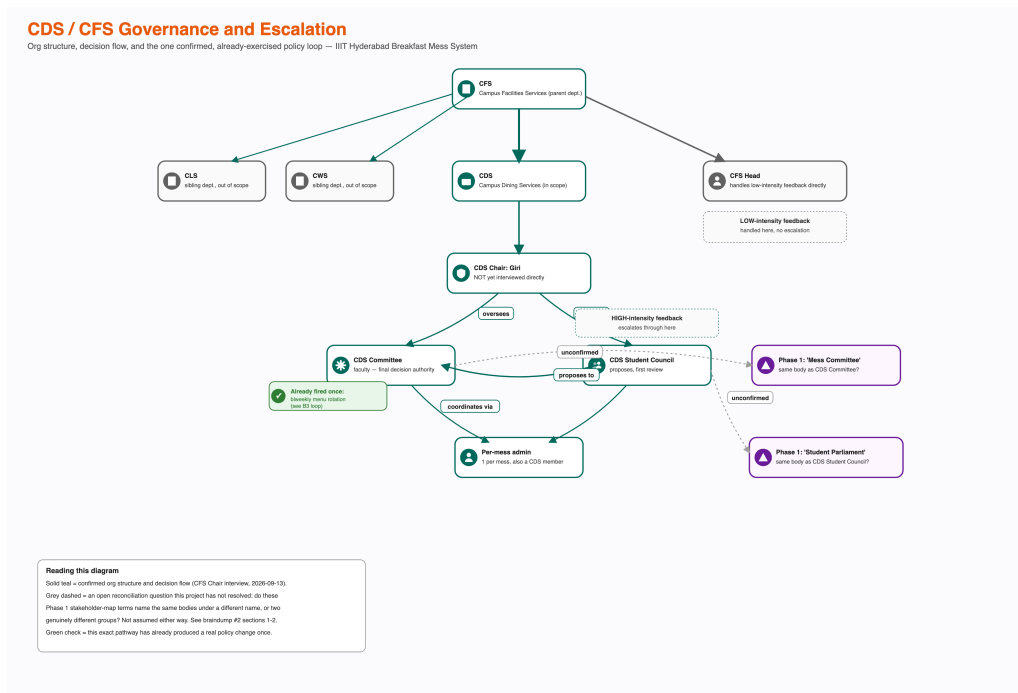


Figure 3: CDS and CFS governance and escalation pathway.

Ten candidate feedback mechanisms were tested. A mechanism counts as a loop only if its last step causally returns to affect its first. Five close; five are real mechanisms that stop partway and are named as such rather than presented as loops.

Feedback Analysis -- Loops, Chains, and Disconfirmed Mechanisms

Closure was tested for every candidate. Only a mechanism whose last variable causally returns to its first counts as a loop.

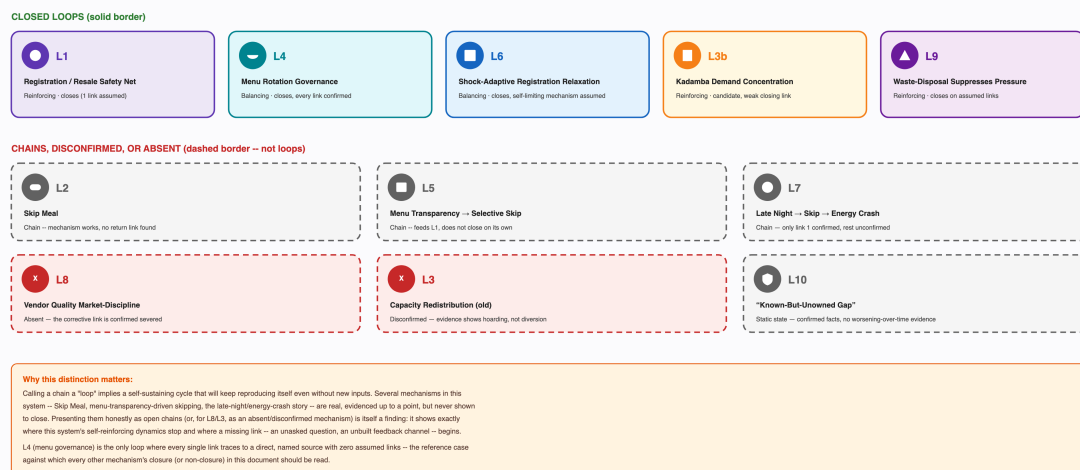


Figure 4: Overview of all tested feedback mechanisms and their closure status.

Loops that close

Registration and resale (reinforcing). Loose registration feeds the Mess Cell resale market; resale removes the financial sting of an unused registration; that softer cost plausibly makes loose registration feel safe again. Most links are confirmed by direct quotes; the final link is a reasonable but unconfirmed assumption.

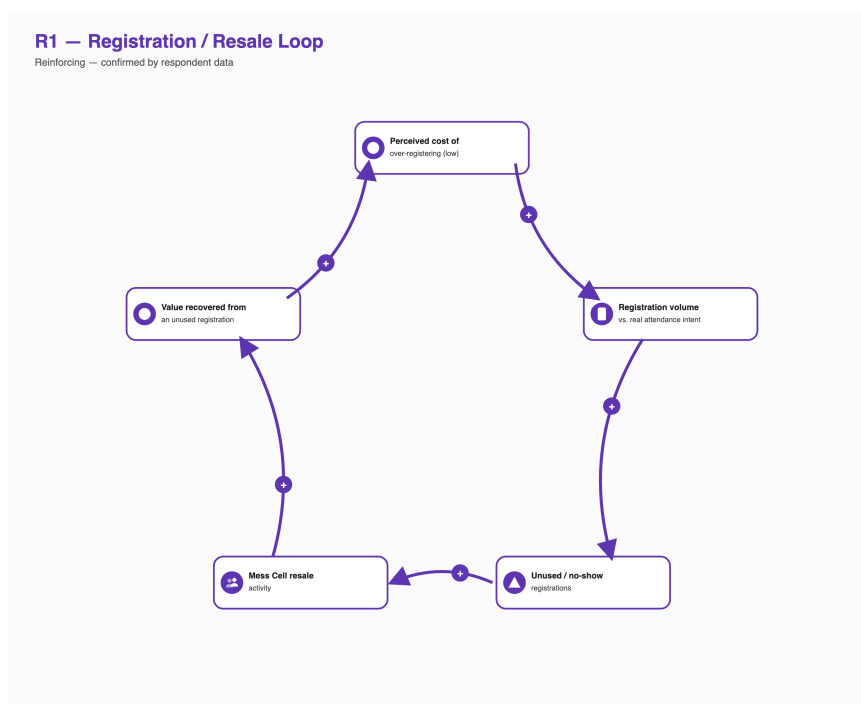


Figure 5: Registration and resale loop.

Menu rotation governance (balancing). Menu fatigue drove feedback, feedback escalated through the Student Council, the Committee approved a rotation, and variety reduced fatigue. Every step confirmed directly by the CFS Chair, the cleanest closed loop in the system.

Shock-adaptive registration relaxation (balancing). Uncertainty rises, registration compulsion relaxes, billing follows real attendance, the vendor absorbs more uncertainty, and confidence in keeping the relaxed model drops, so it reverts. Trigger and mechanism confirmed; whether it closes as a genuine loop or is an on/off switch tied to the calendar remains open.

Kadamba scarcity concentration (reinforcing). Popularity drives defensive month-long registration, which keeps Kadamba scarce. The middle link is confirmed by direct quote; the closing link is weak, since the only available description of a crowded Kadamba is negative.

Waste invisibility (reinforcing). Waste is composted competently enough that it never becomes visible pressure to fix anything upstream. Composting is confirmed; that composting specifically suppresses the pressure is assumed.

Mechanisms that do not close

Mechanism	Why it is not a loop
Skip Meal	Reaches the kitchen and adjusts same-week cooking, but nothing returns to change whether a student uses it. A working one-way mechanism with near-zero uptake.
Menu transparency and selective skipping	A self-identified second-order risk of the rotation fix, with no evidence of anything looping back.
Attendance pressuring vendor quality	The link that would close it is confirmed missing: vendor payment tracks registration, not attendance.
Capacity redistribution to Bakul/Palash	The opposite occurs, scarcity concentrates demand on Kadamba rather than dispersing it.

Mechanism	Why it is not a loop
Known-but-unowned gap worsening over time	Nothing shows it worsening cycle over cycle. A stable stuck state, not a spiral.

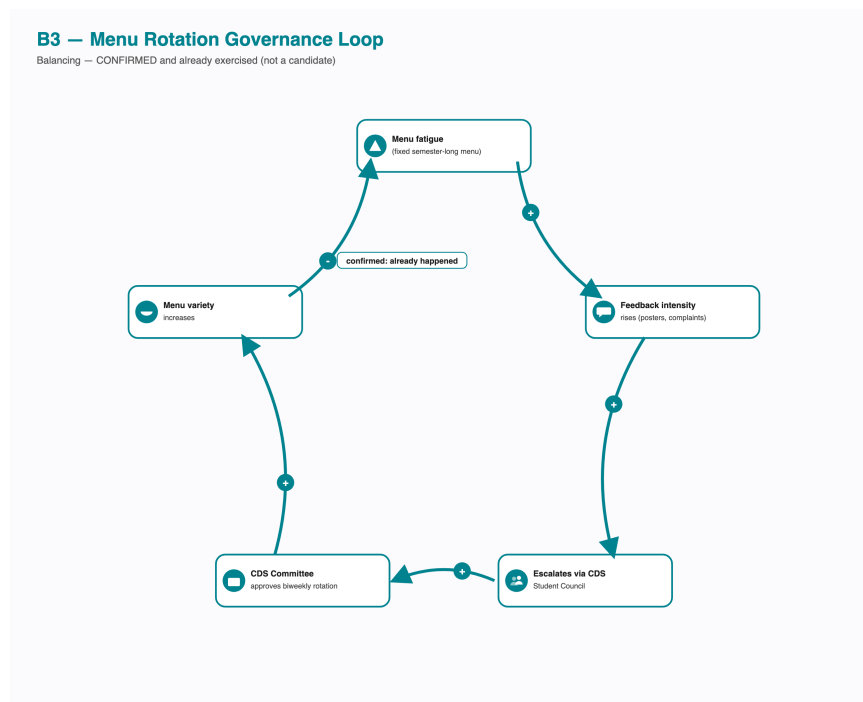


Figure 6: Menu rotation governance loop.

3 Activity 7: Identify Systemic Patterns and Root Causes

Purpose: move from symptoms to the underlying structures and mechanisms producing the problem.

Deliverable 3: Systemic Problem Analysis: patterns, structures, feedback loops and root causes

3.1 The Iceberg Model, six chains

Six areas were traced from event to belief. An event is one thing that happened; a pattern is what keeps happening; a structure is the rule producing the pattern; a mental model is the belief holding the structure up.

Chain	Key structure	Key mental model
1. Register and don't show up	Billing tracks registration, not attendance; Skip Meal returns nothing; walk-in priced 70–90% higher	Registration count is an adequate demand proxy; a registration already made costs nothing to leave unused
2. Food runs out before close	One average consumption figure applied across all items; four-day order lock; one person managing inventory	Procurement certainty matters more than matching food to real demand
3. Complaints ignored until forced	No confirmed committee sign-off required for operational change; four separate decision owners	Operational decisions are the office's alone to make; reducing its own workload justifies a rule change

Chain	Key structure	Key mental model
4. A working fix, switched off	The vendor's four-day lock requires a knowable headcount; nothing exists to evaluate whether a temporary fix should become standing	Attendance-based billing is acceptable only as an emergency measure
5. Menu rotation and its shadow	A real escalation path with authority at the end of it; menu now posted two weeks ahead	Menu variety is worth a policy change; more transparency is automatically good (now being revised)
6. Some of this isn't a problem	Breakfast is a low-companionship meal; only one hall offers separate Jain and pure vegetarian lines	Sleep outweighs a booked breakfast; the system assumes everyone should default into breakfast

3.2 Structural root causes

Three causes recur across the chains, tested against four criteria each: is it structural, does it explain multiple independent patterns, would the pattern collapse without it, and is it evidenced.

RC1, Billing is decoupled from attendance. The financial obligation is created at registration, not consumption. This explains the incentive to over-register, Skip Meal's non-use, why a cancellation cap is needed, why Mess Cell exists, and why a vendor might quietly prefer the current model. Unusually, its counterfactual has actually been run: when billing tracked attendance during the shortage and holidays, the gap closed. Confirmed.

RC2, Decision rights are fragmented. Menu, kitchen execution, academic scheduling and billing policy sit with four separate owners, none holding both the knowledge and the authority. This is why the CFS Chair can state the gap's exact size and it has still never reached a registration or billing policy discussion, and why a proven fix has never been evaluated for standing adoption. Confirmed.

RC3, Attention triggers on complaint intensity, not data severity. Escalation routes high-intensity feedback to the Committee and low-intensity feedback to the CFS Head. Both respond to how loudly a problem is felt, neither to how much it costs. Menu fatigue, sharp, daily, individually felt, travelled the full pathway and produced a policy change. The registration gap, quiet, monthly, diffuse, and already quantified, never has. Mechanism confirmed; the causal generalisation is a candidate.

A fourth, lower-confidence factor: the vendor is paid the same whether or not students attend, so it has little independent reason to push for change. Assumed, pending contract terms.

3.3 Unintended consequences

Rule	Intended function	Unintended consequence
Cancellation cap	Limit uncanceled no-shows	Constrains a behaviour some students don't reliably perform anyway; tightening plausibly pushes traffic into silent no-shows
Walk-in markup	Discourage same-day uncertainty	Universal monthly advance registration, inflating counts independent of intent

Rule	Intended function	Unintended consequence
Skip Meal, no refund	Reduce kitchen waste	Near-zero uptake, so it may not be reducing waste in practice at all
Menu rotation	Fix menu fatigue	Created a way to plan skipping on disliked menu days
Composting and food safety	Keep the mess clean and compliant	Waste never becomes a number anyone must explain, so it cannot create upstream pressure
Outsourcing the kitchen	Save cost, modernise hygiene	Vendor paid on registration regardless of turnout, removing its reason to care about the gap

3.4 System archetypes

Eleven standard archetypes were tested. Five hold: **Shifting the Burden** (resale substituting for a billing fix; composting substituting for measuring waste), **Limits to Growth** (Kadamba's demand hitting a capacity ceiling, answered by building Bakul rather than removing the limit), **Success to the Successful** (Kadamba's scarcity pulling further demand toward itself), **Rule Beating** (registering broadly to guard access, satisfying the rule's letter while defeating its demand-signalling purpose), and **Seeking the Wrong Goal**, the sharpest fit, where the system optimises registration count because it is easy to measure while its own stated goal is minimising waste and matching food to demand.

Two are partial candidates: Fixes That Fail (the December 2024 cap tightening, single-sourced with no before/after data) and Growth and Underinvestment (Bakul/Palash facilities, lacking a confirmed growing-demand precondition). Four are rejected with a stated structural reason each: Tragedy of the Commons, Escalation, Accidental Adversaries and Drifting Goals.

3.5 Systemic tensions

Seven tensions persist because each side is behaving rationally given the structure. Individual rationality against aggregate cost. Procurement certainty against demand-matching accuracy, the sharpest, because the institution has demonstrably operated both. Administrative simplicity against real variation across three different kitchen systems. Transparency against gameability. Loud complaints against quiet costs. A uniform registration model against legitimate cultural, religious and individual variation. And two systems, the academic timetable and the mess window, each solving their own problem with no coordination between them.

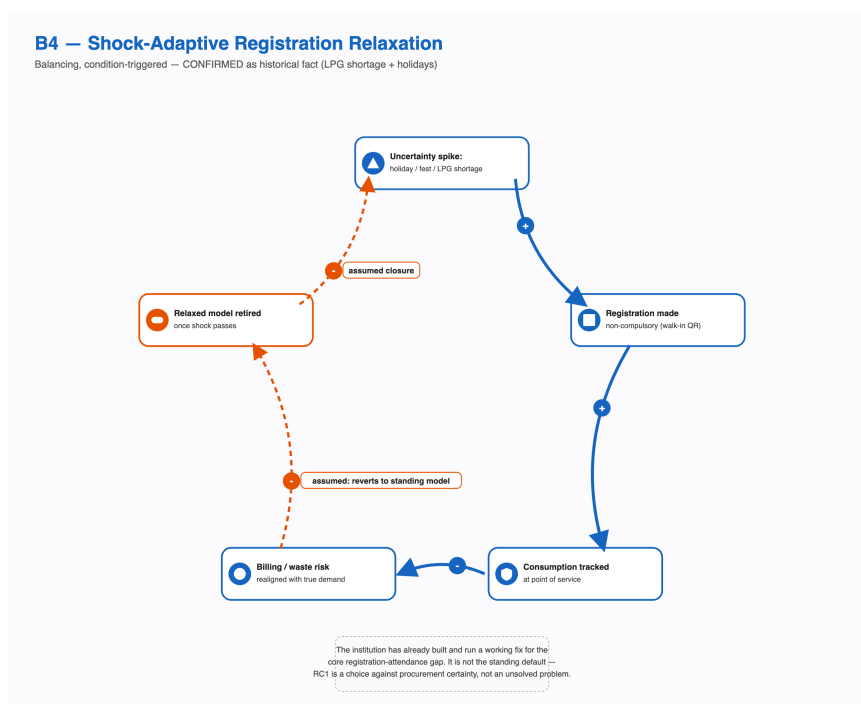


Figure 7: Shock-adaptive registration relaxation loop.

4 Activity 8: Reframe the Problem

Purpose: convert an initial operational problem into a systemic design opportunity.

4.1 Deliverable 4: Reframed Systemic Problem Statement

Deliverable 4: Reframed Systemic Problem Statement

Assumptions challenged

Five assumptions sat inside the original framing, and all five now have evidence against them. Non-attendance is not uniformly a problem, a real share is deliberate fasting, regional habit or religious practice, and the system has no field distinguishing it from an ordinary no-show. Students are not behaving carelessly, every observed decision is rational given the incentives. Nobody has solved this is false, the institution ran attendance-tracked billing twice and it worked. The gap is not unknown, the CFS Chair can state it precisely, and that knowledge has confirmed never reached a policy discussion. And it is not one problem but three structural causes plus one thing that is not a defect.

Stakeholder perspectives compared

Stakeholder	What they experience as the problem	What they do not experience
Students	Being charged Rs. 48 for a meal they cannot use, recoverable only through an informal market	The aggregate, one wasted Rs. 48 never accumulates into something worth escalating

Stakeholder	What they experience as the problem	What they do not experience
Kitchen and serving staff	Cooking to a number they did not set, then facing a 9:20 crowd with popular items gone	Any channel to report what they observe daily
The vendor	Nothing. Registration-based billing insulates revenue from attendance	Any incentive to want this changed (assumed, pending contract terms)
CFS / CDS office	Operational workload, the stated motive behind its one policy change on record	The gap as a live problem; it is a number they hold, not a pressure they feel
CDS Committee	Menu satisfaction, which they successfully fixed	Visibility into what is actually eaten versus registered
Academic administration	Nothing relating to the mess	Any consequence of the 8:30 collision, any channel to mess governance, or any instance of being asked
Faculty and early-shift staff	Undefined, no confirmed access or allocation	Existence, as far as the system's numbers are concerned

The finding this produces is the centre of the reframe: no single actor experiences the Breakfast Paradox as their own problem. The paradox exists only in aggregate, and aggregate is the one thing this system has no mechanism to see, no owner to hold and no channel to escalate.

Why the system is stable rather than broken

The gap persists because every consequence it generates is absorbed before it can become a signal. Mess Cell absorbs the student's financial loss. Composting absorbs the physical surplus. Staff eating unclaimed food absorbs more of it before it is ever counted as waste. Alternative venues absorb the dissatisfaction that would otherwise arrive as a complaint. Informal accommodation absorbs the cultural mismatch that never gets filed as a request.

Each absorption is locally beneficial and none should be removed. Their combined effect is that no financial, volumetric or cultural signal reaches an actor positioned to act. A system with this many competent absorption layers is structurally resistant to generating the pressure a fundamental fix would require.

Alternative framings considered

Framing	Assessment
A, How do we get more registered students to attend?	Rejected. Targets attendance as the goal, colliding with the finding that a real share of non-attendance is legitimate; aims the fix at the actor with least power.
B, How do we make registration a more accurate signal?	Rejected. Assumes better student behaviour is the missing ingredient. The institution's own proven alternative required no student behaviour change at all.
C, Who should own the registration-attendance gap?	Closer but incomplete. Explains why a known fix was never picked up, not why the gap never became loud enough to demand an owner.

Framing	Assessment
D, The systemic framing	Adopted. The gap is a stable equilibrium maintained by three structures and insulated by locally beneficial absorption mechanisms.

Validation

Framing D is evidenced at each joint. That the fix is possible is not hypothesis, it ran twice and closed the gap, the strongest form a counterfactual can take. That fragmentation is why it was not kept is confirmed by the CFS Chair's knowledge never reaching policy. That attention is intensity-triggered is evidenced by one escalation design producing two opposite outcomes. One joint is unvalidated and named as such: whether the vendor's contract could absorb a standing version of the attendance-tracked model.

The Breakfast Paradox is not a gap between registration and attendance. It is a stable equilibrium the system maintains around that gap, because every actor's behaviour within it is individually rational and every consequence it produces is absorbed before it can become a signal.

Billing fires at registration rather than at consumption. That single structure removes the cost of inaccuracy from the student and the revenue risk from the vendor at the same time, so neither party closest to the daily decision has reason to want the number to be accurate. What follows is quietly absorbed: the resale market takes the financial loss, composting and staff consumption take the surplus, outside venues take the dissatisfaction, informal workarounds take the cultural mismatch. Each absorption is locally useful and none should be removed. Together they ensure the gap never generates a signal loud enough to reach a system whose attention is triggered by complaint intensity rather than measured cost, and whose decision rights over this outcome are split across four actors none of whom holds both the knowledge and the authority to act.

This is why the one office that can state the gap's exact size has never had reason to raise it, and why an attendance-tracked model that already worked twice was retired both times without evaluation. No single actor experiences this as their own problem, the student experiences a charge, the kitchen a rush, the office a statistic, the vendor nothing at all. The paradox exists only in aggregate, and aggregate is the one thing this system has no mechanism to see.

4.2 Deliverable 5: Design Opportunity Statement

Deliverable 5: Design Opportunity Statement

Where the leverage is

Not at the event layer, persuading individual students to attend targets the most visible layer and the least movable one. Not at the parameter layer either, which is where the institution acts when it acts: the cancellation cap has already moved from five to ten and back, changing where pressure escapes rather than whether it exists. The leverage is at the structural layer, what triggers a bill, and what triggers institutional attention, and beneath that at the mental-model layer, in the belief that registration count adequately stands in for demand and that

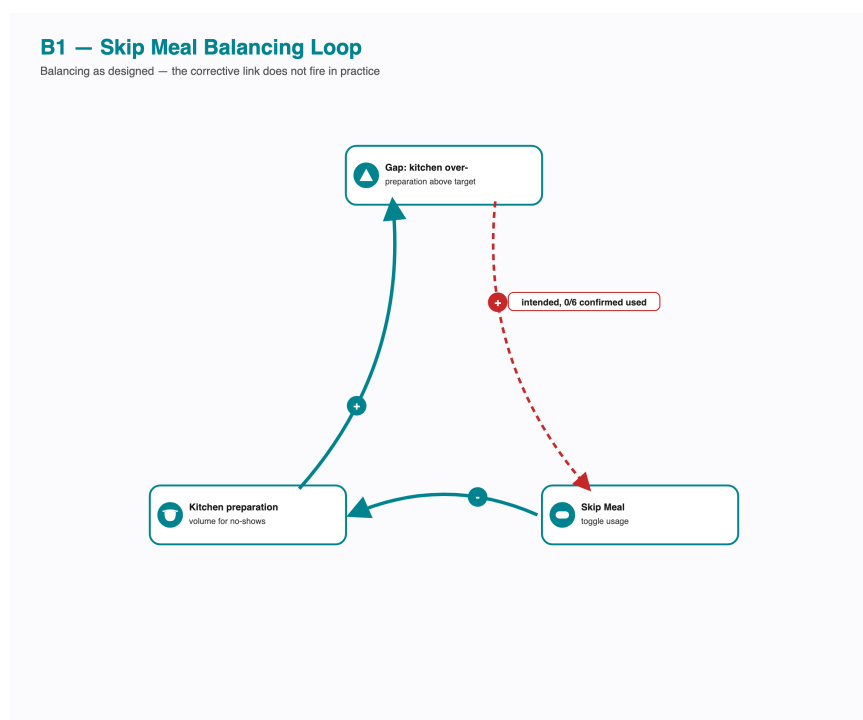


Figure 8: Skip Meal, a functioning one-way mechanism rather than a closed loop.

procurement certainty permanently outweighs demand accuracy. The institution has already suspended both beliefs twice, so neither is as fixed as it appears.

The opportunity is to give this system a way to notice its own cost.

The Breakfast Paradox persists not because the fix is unknown, it has already been run twice, but because nothing converts the cost of the gap into a signal reaching anyone with authority to act. Every consequence is absorbed competently before it becomes visible, and the institution's attention triggers on complaint intensity, while this problem is quiet. The design opportunity is therefore not to build a new solution but to build the missing feedback: a route by which a known, measured, aggregate cost can travel upward and become a decision.

Three intervention points

Layer	Opportunity
The rule (RC1)	A standing, partial version of the attendance-tracked model the institution has already proven, scoped to breakfast specifically, worst turnout, least procurement volume at risk. Gated on what the vendor's contract can absorb.
Decision rights (RC2)	A single named owner accountable for the registration-attendance outcome end to end. Not a new committee, the existing escalation structures work when something reaches them. What is missing is an actor for whom this gap is their problem.

Layer	Opportunity
Information flow (RC3)	An escalation trigger firing on measured cost alongside the existing one firing on complaint volume. The cheapest of the three and the broadest in reach. Its precondition is measuring waste by volume, which is currently never done.

Constraints any intervention must respect

Do not remove an absorption mechanism to manufacture a signal, making resale harder would make real students worse off to create visibility the institution could measure directly. Do not treat turnout as the success metric, since a share of the gap is people eating the way they actually live. Do not treat the 8:30 class start as the lever; the academic timetable solves real problems of its own and should be treated as a boundary condition. Do not aim the fix at students, who hold the least power and the most defensible behaviour. And preserve what already works, the escalation pathway and Skip Meal's kitchen mechanism both function and need extending, not replacing.

Sequencing

The three are not alternatives. Measuring cost is the enabling step and can be done unilaterally by an office already holding the data. Assigning an owner converts a visible number into a decision, a signal with no addressee produces nothing, which is precisely the current state. Changing the billing rule is deepest and gated externally, best understood as what the first two make possible.

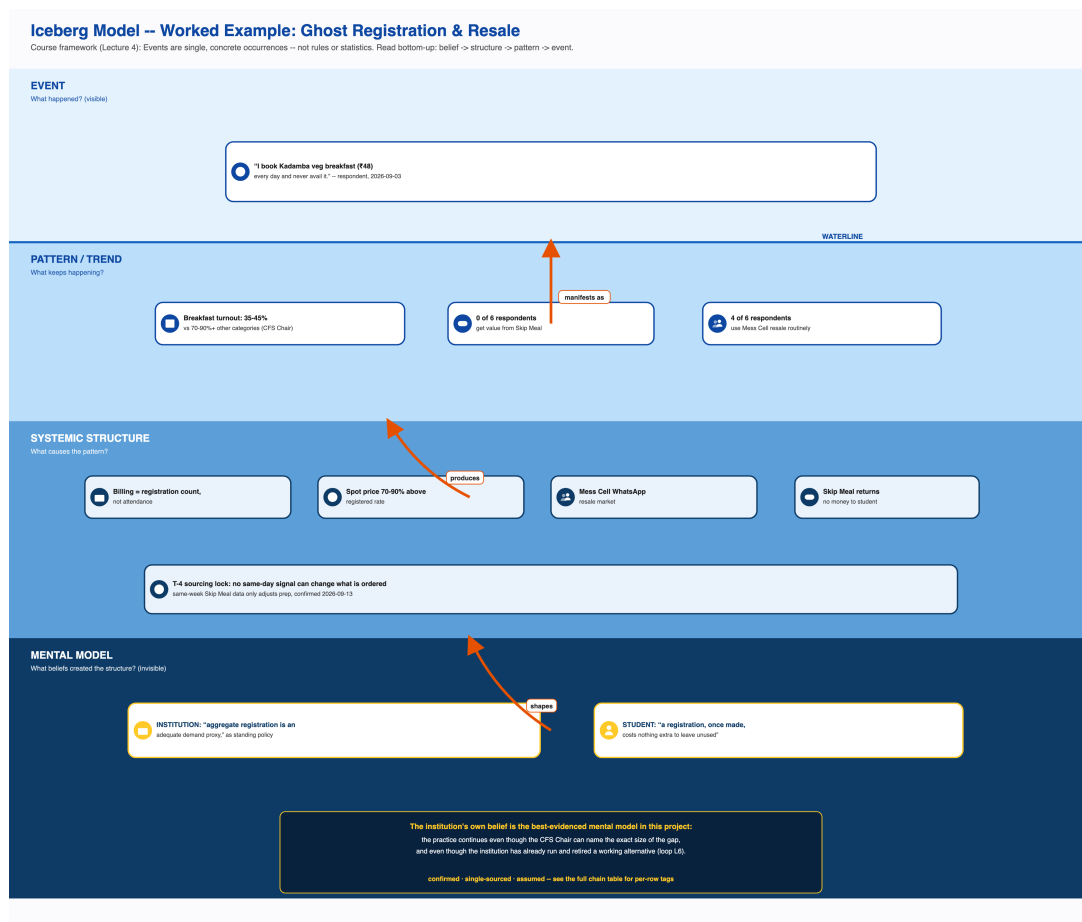


Figure 9: Iceberg Model across the six traced chains.

5 Outstanding Data Collection

The following remain unresolved and are stated as open rather than assumed past.

- **Whether the vendor's contract could absorb a standing or partial attendance-tracked model**, and if not, which specific term blocks it. Never asked directly. This determines whether the deepest intervention is an adoption problem or a renegotiation problem, and is the single highest-priority open item.
- **Why Skip Meal usage is near zero** despite its kitchen mechanism being confirmed to work. An awareness gap, a trust gap and a portal-friction problem each imply a different fix.
- **An interview with the CDS Chair**, the actor with the most formal authority in the dining chain and the only one never interviewed. Every account of CDS decision-making here comes from one level up.
- **Whether CDS/CDS Committee/CDS Student Council are the same bodies** earlier phases called Mess Office, Mess Committee and Student Parliament, and whether Warden and CFS are the same role.
- **Kadamba's two capacity figures** (1200 registered versus 420 seats) and which of the two portal addresses on the institution's own posters is authoritative.
- **Direct observation of a live service window**, still not conducted; a 30-minute interval protocol across 7:30–9:30 AM is prepared.

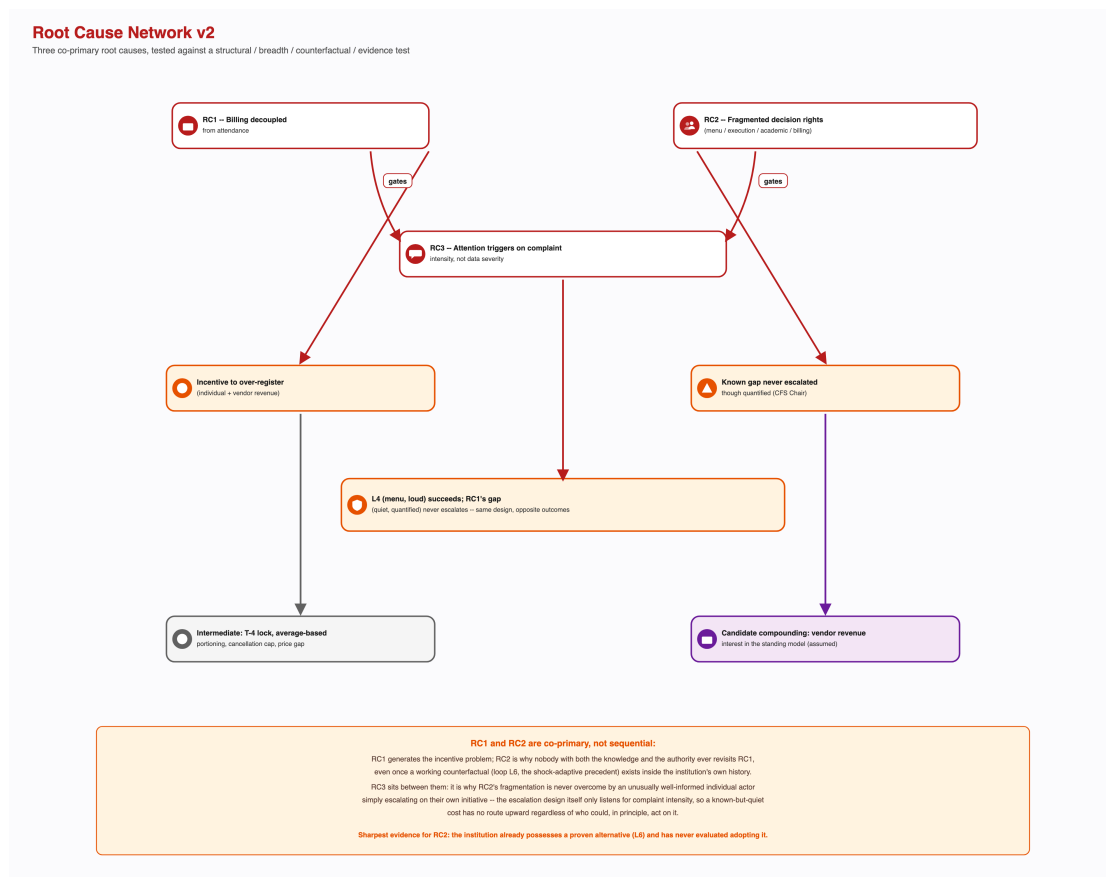


Figure 10: Root cause network, how RC1, RC2 and RC3 interact and compound.

- **Whether faculty and early-shift staff have any real breakfast access**, which if so is a second demand stream invisible in every number the system tracks.

Phase 2 covers Activities 6, 7 and 8 of the Generic Systems Thinking & Design Project Framework, and the five deliverables those activities produce. Section structure follows the framework's own activity and deliverable ordering.